



AAE451DBF: A3

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DELIVERABLE 1 - ASSUMPTIONS

Battery Density: 5.27×10^5 J/kg

- Given in 04 notes. Matches standard values for Li-Po batteries

Motor Efficiency (η_m): 0.7

- Calculated from Cobra Motor Datasheet to be 0.8-0.85, chose 0.7 to be conservative prior to final motor choice.

Empty Weight Fraction Estimate (W_e): 0.8

- Assume weight fraction on higher end due to lack of DBF experience on team, provide extra margin.

Usable Battery Capacity (80%)

- Taken from Lecture slides.

Temperature Derate (70%)

- 70% efficiency, assuming historical weather data based on Li-Po battery temperature charts.

All other assumptions are the same as in A2.

DELIVERABLE 2 – MISSION PROFILE & ENERGY ESTIMATE

1. Warm Up:

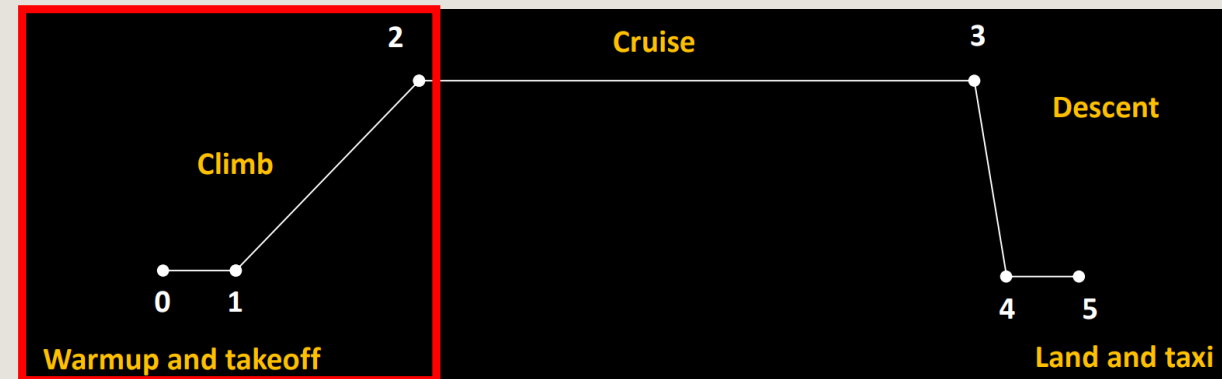
- Equation used: $W_{WU}^B = N \frac{W_{TO}^B}{W}$
- Output: 0.0365

1. Take Off:

- Equation used: $W_{TO}^B = \frac{T_{TO}}{\eta_m \left(\frac{W}{P}\right) \rho_B}$
- Output: 0.0037

2. Climb:

- Equation used: $W_{CL}^B = \frac{VT_{CL}}{\eta_p \eta_m \rho_B} \left(\frac{\cos \gamma}{L/D} + \sin \gamma \right)$
- Output: 0.0017



DELIVERABLE 2 – MISSION PROFILE & ENERGY ESTIMATE

4. Cruise:

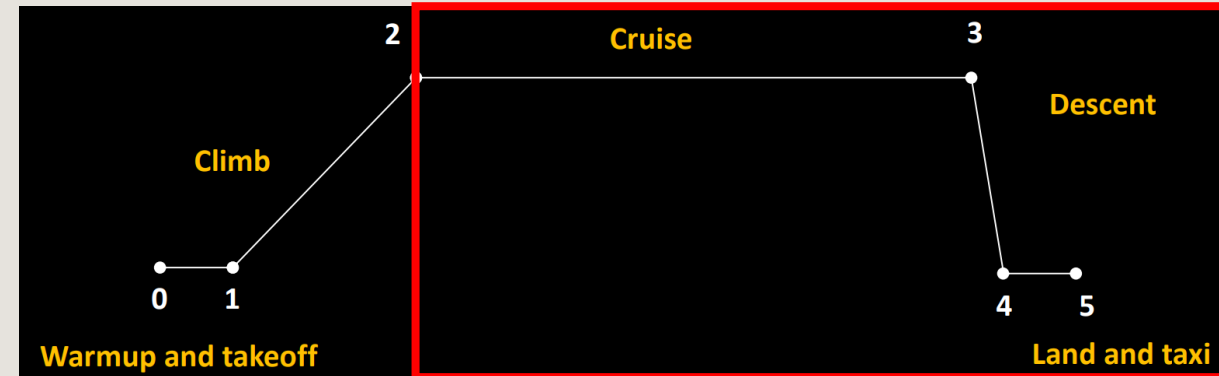
- Equation used: $W_{LF}^B = \frac{VT_{LF}}{L/D\eta_p\eta_m\rho_B}$
- Output: 0.0027

5. Descent: Assumed to be negligible

- Equation used: N/A
- Output: 0

6. Land & Taxi: Assumed to be negligible

- Equation used: N/A
- Output: 0
- Energy Margin Calculated:**
 - Equation used:
 - Output: 1.7857



$$E_{\text{margin}} = \frac{E_{\text{Batt Energy}}}{E_{\text{Required by Plane}}}$$

DELIVERABLE 3 – EMPTY WEIGHT FRACTION

Empty Weight Fraction Estimate: 0.8

- Rationale: Assume weight fraction on higher end due to lack of DBF experience on team, provide extra margin

Weight Range Verification: ■ Team 1 Estimate



DELIVERABLE 4 - MOTW

Equations Used:

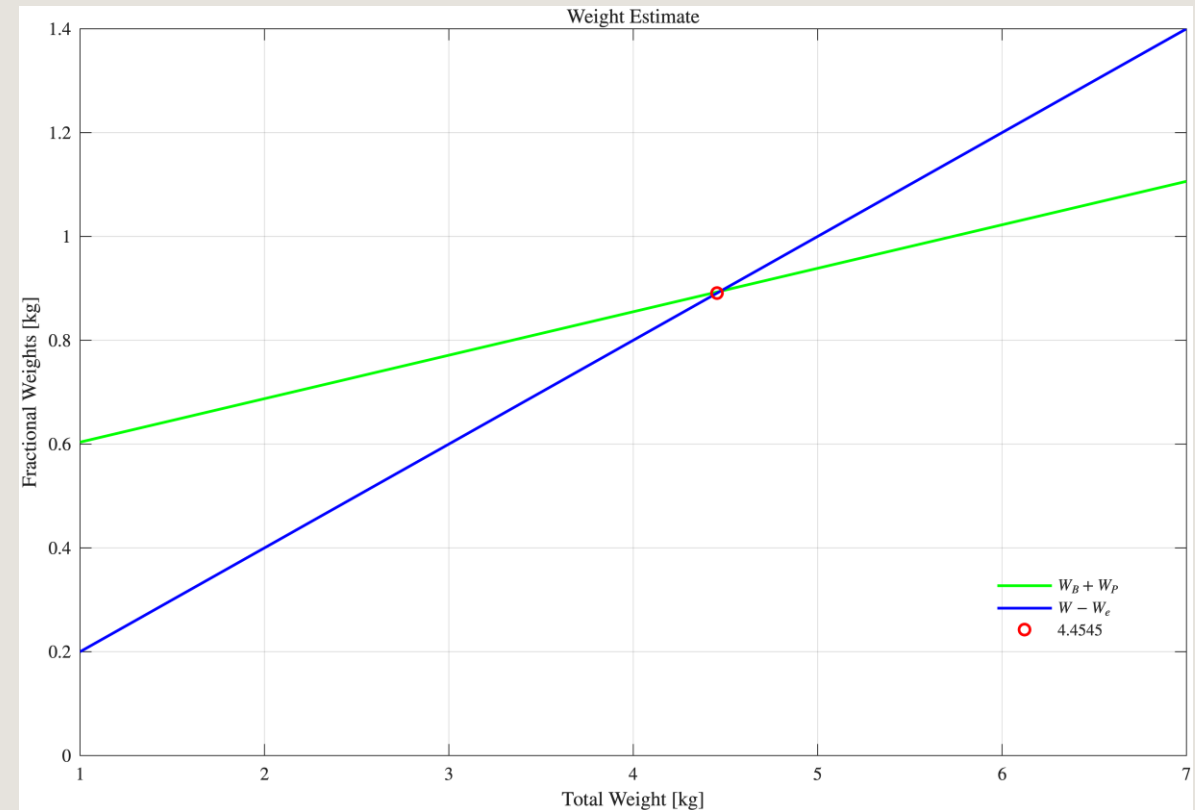
1. Weight of the battery is dependent on the total weight, so we vary over a range of values:

$$(W^B) \cdot W + W_{\text{pay}}$$

2. The historical weight estimate relies on the total weight and historical empty weight fraction values:

$$W - W_e$$

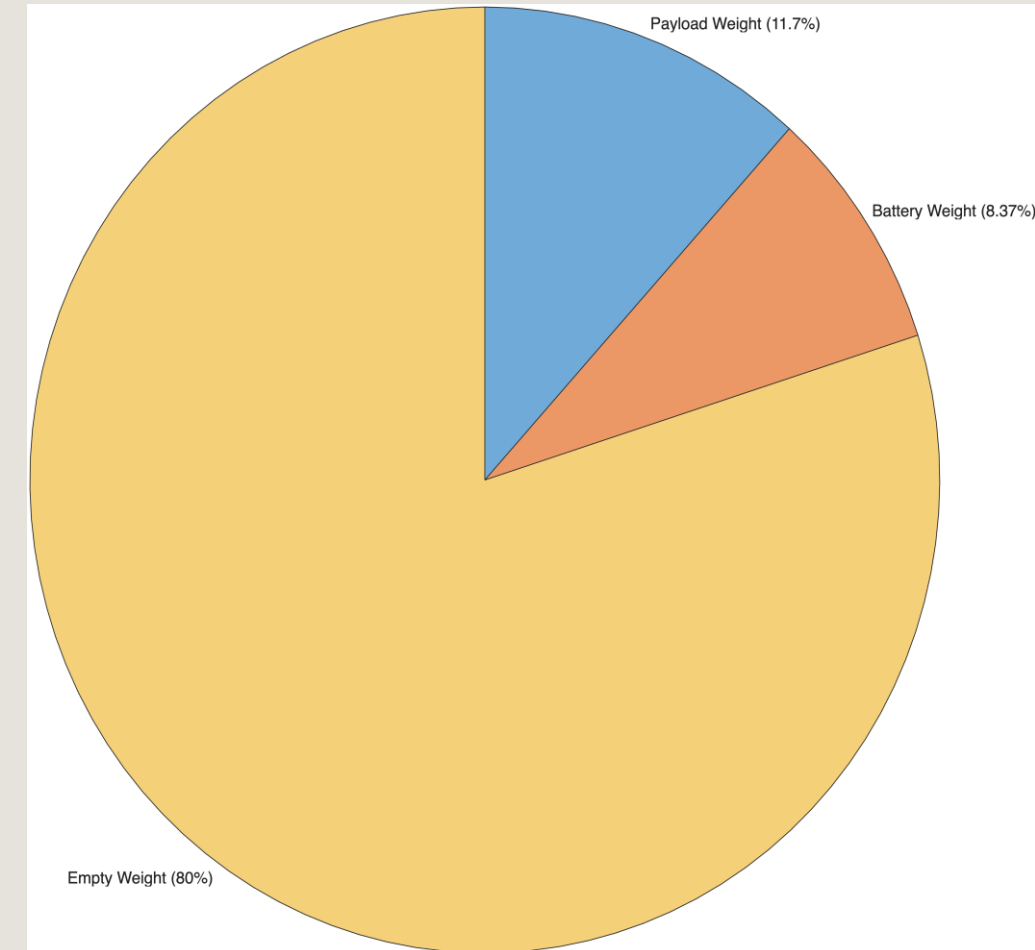
Final weight estimate (4.45 kg) is the intersection of these two lines, marked in red on the plot.



DELIVERABLE 5 – WEIGHT BUILDUP TABLE

- 80% of weight for empty weight, based on our assumption of the empty weight fraction
- 11.7% of the remaining weight is payload weight of 0.52 kg.
- Last 8.37% of the weight is the battery weight (0.373 kg)

Weight Table	
Weight Allocation	Weight (kg)
W_e	3.56
W_b	0.37
W_p	0.52



DELIVERABLE 6 – TOTAL ENERGY CALCULATION

1. Actual Energy Required per Flight Phase:

1. Warm Up: 85.7 kJ
2. Take Off: 8.6 kJ
3. Climb: 4.04 kJ
4. Cruise: 5.4 kJ
5. Descent: 0 J
6. Land & Taxi: 0 J

2. How Much Energy is Lost:

- 92.2 kJ

3. Total Energy Required by Battery:

- 197 kJ